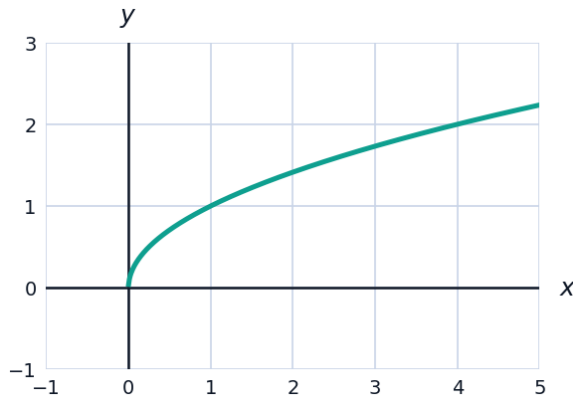


Area Between Curves



Area between two curves

- 1 Find the area of the region bounded by $y = x^2$ and $y = x + 2$.
- 2 The region bounded by $y = \sqrt{x}$, $y = 0$, and $x = 4$ is shown. Find its area.



Setting up with respect to y , and total area

- 3 Find the area enclosed by $x = y^2$ and $x = 2y$ by integrating with respect to y .
- 4 Find the total area between $f(x) = \sin x$ and the x -axis on $[0, 2\pi]$ (note the curve is below the axis on part of this interval).